Lab 05

Designing Test Cases for Different Software Testing Categories

Part A – Identify Test Scenarios

Task A1: Identify Core Scenarios

- Selected Test Scenarios
    1. User Registration
       New users creating accounts with valid and invalid inputs.
    2. User Login / Authentication
       Verification of valid and invalid credentials.
    3. Product Browsing and Search
       Viewing products by category and retrieving correct results.
    4. Add to Cart Functionality
       Adding and removing products from the shopping cart.
    5. Cart Calculation
       Accurate total price calculation including quantity updates.
    6. Order Placement
       Creating an order from cart items.
    7. Online Payment Processing
       Secure and successful transaction handling.
    8. Admin Product Management
       Adding, updating, and deleting product information.

9. Session Management
   Proper login session handling and logout functionality.
10. System Response Under Load
    Behaviour when multiple users access the system simultaneously.

Part B – Write Test Cases by Category

1. Sanity Testing

Purpose

To verify that critical functionalities work after a build or change.

- ST-1: User Registration – Valid Input

Test Case ID: ST-1
Objective: Verify user can register with valid details
Precondition: User is on Registration Page
Test Steps:

1. Enter valid name, email, and password
2. Click “Register”

Expected Result:
User account is successfully created and confirmation message is displayed.

- ST-2: User Login – Valid Credentials

Test Case ID: ST-2
Objective: Verify user can log in with correct credentials
Precondition: User account already exists
Test Steps:

1. Enter valid email and password

2. Click “Login”

Expected Result:
User is redirected to dashboard/home page.

- ST-3: Browse Products

Test Case ID: ST-3
Objective: Verify products are displayed properly
Precondition: Products exist in the system
Test Steps:

1. Navigate to product listing page
2. Select a category

Expected Result:
Products under the selected category are displayed correctly.

- ST-4: Add Product to Cart

Test Case ID: ST-4
Objective: Verify product can be added to cart
Precondition: User is logged in
Test Steps:

1. Select a product
2. Click “Add to Cart”

Expected Result:
Product is added to cart and cart count updates.

- ST-5: Place Order

Test Case ID: ST-5
Objective: Verify user can place an order
Precondition: Product exists in cart
Test Steps:

1. Navigate to cart
2. Click “Checkout”
3. Confirm order

Expected Result:
Order is successfully created and confirmation message is displayed.

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2. Security Testing

Purpose

To ensure the system protects data and prevents unauthorized access.

- SEC-1: Invalid Login Attempt

Test Case ID: SEC-1
Objective: Verify system rejects invalid login credentials
Precondition: User is on Login Page
Test Steps:

1. Enter incorrect email/password
2. Click “Login”

Expected Result:
System displays error message and denies access.

- SEC-2: Access Admin Page Without Authorization

Test Case ID: SEC-2
Objective: Verify unauthorized users cannot access admin functionality
Precondition: User is logged in as normal customer
Test Steps:

1. Attempt to access Admin URL directly

Expected Result:
Access is denied and user is redirected or shown an authorization error.

- SEC-3: SQL Injection Attempt

Test Case ID: SEC-3
Objective: Verify system prevents SQL injection attacks
Precondition: User is on Login Page
Test Steps:

1. Enter ' OR '1'='1 in username field
2. Enter random password
3. Click “Login”

Expected Result:
Login fails and input is properly sanitized.

- SEC-4: Password Storage Protection

Test Case ID: SEC-4
Objective: Verify passwords are not stored in plain text
Precondition: User account exists
Test Steps:

1. Check stored password format in database

Expected Result:
Password is stored in encrypted/hashed format.

- SEC-5: Session Timeout Validation

Test Case ID: SEC-5
Objective: Verify session expires after inactivity
Precondition: User is logged in
Test Steps:

1. Stay inactive for predefined session timeout period
2. Attempt to perform any action

Expected Result:
User is automatically logged out and required to log in again.

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3. Stress Testing

Purpose

To evaluate system behaviour under extreme load conditions.

- STR-1: High Concurrent User Load

Test Case ID: STR-1
Objective: Evaluate system performance under a high number of simultaneous users
Precondition: System is deployed and running
Test Steps:

1. Simulate 500–1000 concurrent users accessing the website
2. Perform actions such as login and browsing

Expected Result:
System may slow down but should not crash.
System should handle requests gracefully or show controlled error messages.

- STR-2: High Volume Order Processing

Test Case ID: STR-2
Objective: Test system behavior when processing a large number of orders simultaneously
Precondition: Products exist in database
Test Steps:

1. Simulate multiple users placing orders at the same time
2. Monitor order creation and database updates

Expected Result:
Orders are processed without data corruption or duplication.
System maintains data consistency.

- STR-3: Large Database Load

Test Case ID: STR-3
Objective: Evaluate system performance with a large product database
Precondition: Database contains 10,000+ products
Test Steps:

1. Load product listing page
2. Perform product search and filtering

Expected Result:
System should respond within acceptable time limits or degrade gracefully without crashing.

- STR-4: Limited Server Resources

Test Case ID: STR-4
Objective: Evaluate system behavior under reduced CPU or memory availability
Precondition: Server resource usage is intentionally limited
Test Steps:

1. Simulate normal user traffic
2. Monitor response and system stability

Expected Result:
System may experience performance degradation but should not terminate unexpectedly.
Proper error handling should be observed.

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4. Performance Testing

Purpose

To verify response time, throughput, and resource usage.

- PERF-1: Page Response Time

Test Case ID: PERF-1
Objective: Verify that core pages load within acceptable response time
Precondition: System is deployed and stable
Test Steps:

1. Access Home Page
2. Access Product Listing Page
3. Access Cart Page

Expected Result:
Each page should load within 2 seconds for normal user traffic.

- PERF-2: Login Response Under Normal Load

Test Case ID: PERF-2
Objective: Verify login response time with concurrent users
Precondition: 100 concurrent users simulated
Test Steps:

1. Simulate 100 users attempting login simultaneously
2. Measure response time

Expected Result:
Login response time should remain within acceptable limits (e.g., under 3 seconds).
No request should fail.

- PERF-3: Order Processing Throughput

Test Case ID: PERF-3
Objective: Measure number of orders processed per minute
Precondition: System running with stable database
Test Steps:

1. Simulate multiple users placing orders
2. Monitor number of successful transactions per minute

Expected Result:
System should process expected transaction volume without delay or data inconsistency.

- PERF-4: Scalability Testing

Test Case ID: PERF-4
Objective: Evaluate system scalability as user load increases
Precondition: System running on test server
Test Steps:

1. Gradually increase users from 50 to 200
2. Monitor CPU, memory usage, and response time

Expected Result:
System performance should degrade gradually and remain stable without crashing.
Resource usage should scale predictably.

Part C – Review & Reflection

Task C1: Validation Questions

1. Are all test cases traceable to SRS/use cases?

Yes. Each test case is derived from the functional and non-functional requirements specified in the SRS and the identified UML use cases.

- Sanity test cases validate core use cases such as Registration, Login, Browse Products, Add to Cart, and Place Order.
- Security test cases trace back to authentication and authorization requirements.
- Stress and performance test cases correspond to non-functional requirements related to scalability, response time, and system stability.

Therefore, all test cases are directly traceable to defined system requirements.

2. Which testing category was hardest to write? Why?

- Stress testing was the hardest to write because it requires defining extreme load conditions and predicting system behavior under unusual scenarios. Unlike functional testing, stress testing involves estimating system limits, resource constraints, and concurrency behavior, which are

not always explicitly defined in the SRS. It requires deeper understanding of system architecture and infrastructure.

3. Which defects would not be caught by sanity testing?

Sanity testing only verifies basic functionality and workflow. It would not catch:

- Security vulnerabilities such as SQL injection or unauthorized access
- Performance bottlenecks under high load
- Memory leaks or resource exhaustion issues
- Data integrity issues under concurrent access
- Session management weaknesses

These types of defects require specialized testing such as security, stress, or performance testing.

Conclusion

In this lab, various software testing categories were applied to the E-Commerce Website to evaluate its correctness, security, stability, and performance. Core scenarios were identified from the SRS and UML models, ensuring that all test cases were traceable to defined requirements. Sanity testing verified basic workflow functionality, while security testing ensured protection against unauthorized access and vulnerabilities. Stress testing evaluated system behavior under extreme conditions, and performance testing measured response time, scalability, and resource usage.

This lab provided practical understanding of how different testing approaches complement each other to ensure software quality. It demonstrated that functional correctness alone is not sufficient; a reliable system must also be secure, scalable, and robust under real-world conditions.

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